

Center for Independent Experts (CIE) Independent Peer Review of the Alaska-wide Sablefish Stock Assessment

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Contents

Contents	2
1. Executive Summary.....	3
2. Review Process.....	4
3. Terms of Reference 1	5
ToR 1 Recommendations/Findings	8
4. Terms of Reference 2	9
ToR 2 Recommendations/Findings	12
5. Terms of Reference 3	13
ToR 3 Recommendations/Findings	14
6. Appendix 1: Bibliography	16
7. Appendix 2: Performance work statement	17
8. Appendix 3: Panel attendance	26

1. Executive Summary

Excellent background information was provided for the review and comprehensive presentations and discussion were held. I was impressed by the quality of work that has been conducted, including the quality of the codebase and the extent to which the work has been published in peer reviewed journals.

Changes to fishing and the scientific survey have introduced new challenges for the existing single region panmictic assessment. The presented fleets as areas (FAA) model addresses these issues whilst avoiding the complexity of a full spatial assessment model. A full spatial assessment model was also presented but uncertainty in stock structure and input data limit its additional utility.

The scientific survey is well structured and implemented. However, the assessment is limited by the scale of the survey, the extent of observer coverage, the lack of catch and effort logbooks and the number of fish that are aged. Within these constraints, the FAA model, 26.2_FAA, provided a good fit to the data and provides the best balance between the model complexity and capturing sufficient biological detail.

From the presented diagnostics and discussions 26.2_FAA is appropriate for assessing stock status and supporting management decisions. The resulting OFL and ABC are lower than for the single region panmictic assessment, however the difference is not of unexpected magnitude for a significant model change. Overall, use of this model is consistent with using the best scientific information available (BSIA).

A spatial assessment model was also presented. The data and understanding of population structure needed to support this model was limited; this is akin to the limitations observed with more complex FAA models. A key finding consistent across the presented models is that uncertainty in stock structure and data limitations are restricting the application of more complex models.

The results and discussions revealed a number of potential research directions that would support the future development of a spatial model (or improved FAA model). Prioritization of this work would have to consider the relative costs and benefits (out of scope of this review).

2. Background

Sablefish (*Anoplopoma fimbria*) are a long-lived (90+ years), highly mobile, sexually dimorphic demersal species that live at depths greater than 200m (as adults). In Alaska Federal waters, sablefish are assessed and managed as a single population with the total quota apportioned across 6 management regions (Aleutian Islands, Bering Sea, Western Gulf of Alaska, Central Gulf of Alaska, Western Yakutat, and Southeast Outside) and among two fishing sectors (i.e., fixed gear and trawl gear). Since rationalization (i.e., implementation of individual fishing quotas, IFQs) occurred in the 1990s, sablefish have been one of the most valuable finfish resources in Alaska (\$124 million in ex-vessel value in 2022). Catch advice for sablefish is based on a statistical catch-at-age integrated stock assessment model fit to fishery-independent survey data and fishery-dependent catch-age data. The sablefish assessment is reviewed annually by the North Pacific Fishery Management Council's (NPFMC) Science and Statistical Committee (SSC) as part of the quota specification process, and last underwent a CIE review in 2016.

3. Review Process and Role of Reviewer

The CIE review for the Sablefish Stock in Alaska took place over three days in Juneau from 16-18 June 2026. The meeting was chaired by Aaron Berger, with CIE reviewers Peter Stephenson, Sam Subbey and Klaas Hartmann (myself).

Comprehensive documentation was provided in advance, this included:

- A review document written specifically for the CIE review with comprehensive details (178pp., (Goethel and Cheng 2026))
- Five appendices that focus on specific issues to investigate during the review (392pp.)
- A github repository with code and model runs for most of the results provided in the main report and appendices
- Presentations for most of the sessions covered in the three days of the review
- Previous (2024 and 2021) assessment reports
- Extensive supporting documents categorized by topic (131 documents, mostly journal publications)

The quality and breadth of the background documents enabled the team to be well prepared prior to the site visit. The CIE review document (and appendices) was excellent and was key to enabling me to efficiently familiarize myself with the fishery and be able to concentrate on key topics and the terms of reference. I was also impressed with the amount of work on this assessment that has been published in the peer reviewed literature.

The extensive code repository on github was well organized with good documentation and relatively straightforward to install. The assessments were easy to modify and re-run. The stock assessment team were able to efficiently conduct analyses requested by the CIE reviewers. Consequently, limited analyses were conducted by the CIE reviewers themselves; I ran only a small number of analyses to confirm my understanding of the model and conduct some basic exploration.

The site visit was highly organized with a clear agenda. The agenda was well structured with an appropriate mix of presentations and opportunity for discussion. The required experts on key topics were present during appropriate topics and available as further questions arose. Several additional analyses were requested by the CIE team, including myself, and these were readily conducted with sufficient time for discussing the results throughout the meeting.

4. Terms of Reference 1

Review the disseminated documents and analyses, then provide feedback on whether the fleets-as-areas (FAA) assessment provides a suitable basis for operational management advice, particularly in comparison to the current single region panmictic assessment.

a) Provide recommendations on the most appropriate parametrization of fleet structure (number of fleets per fishery/survey, as a proxy for regions) in the FAA model.

The assessment has previously been conducted using a single area stock assessment model that was developed specifically for this fishery (e.g., (Goethel and Cheng 2024)). Genetic studies indicate a genetic panmixia across the broader region (Jasonowicz et al. 2017; Timm et al. 2024). However, Sablefish are long lived and it is understood that mixing rates within Alaska are low with periodic migrations occurring. Factors influencing migration are poorly understood, however evidence indicates movement is age specific with a general pattern of juvenile fish moving from the Gulf of Alaska (GoA) to the Bering Sea (BS) and Aleutian Islands (AI) whilst mature fish move Eastwards to the GoA.

Typical movement between the management areas is variable with estimated annual movement rates out of any given area ranging between 10% and 88% (Hanselman et al. 2015). A broader spatial study found that annual retention rates within the Alaskan stock were 92% (Kapur et al. 2025). However, this considered only movement down the North American coast, not movement across the Western boundary to Kamchatka.

Given the biological spatial structure, use of a single region model can be inappropriate, particularly if the spatial distribution of catches and/or the spatial coverage of scientific

data is not comprehensive or changes through time. Both of these changes have occurred in recent years:

Spatial catch distribution

The proportional TAC distribution has historically ensured a consistent proportion of catches across the regions, however recent undercatch of TAC has enabled the proportions to change. Since 2019 the proportion of catch from the Bering Sea has almost doubled while proportions in the Central and Eastern GOA have decreased by 25%.

Changes in age frequency data are apparent that are too rapid to be explained by demographic processes in a well-mixed population (e.g., rapid emergence and disappearance of a plus group cohort). These were most apparent in observer data and may have been due to spatial shifts in the fishing locations. However, some of these trends were also present in the scientific survey.

Catch and effort logbooks are unavailable, consequently spatial shifts in catch within each of the five management areas are not recorded or understood. Although additional analysis of observer data may be possible.

Scientific data changes

The fishery independent survey does not cover all regions in each year. Previously the GoA has been sampled every year with BS and AI sampled in alternating years. Changing economic conditions and vessel availability have altered the sampling regime such that GoA is sampled every second year and the BS/AI sampled together in every second year.

Relative Population Numbers (RPN)

For the single region model a single abundance index, referred to as the relative population numbers (RPN) is required. This needs to be derived from the scientific index which has varying spatial coverage.

Previously, in each year either AI or BS was not sampled. The missing RPN for this region was extrapolated by assuming the RPN had the same relative change as GOA. The regional RPNs were then combined to provide the single RPN for the model.

Juveniles are proportionally more abundant in the AI/BS than in the GOA. Consequently, with this method, following a recruitment pulse, the GOA RPN will potentially lag behind AI/BS and the reverse in a period of pronounced recruitment reduction. The extrapolation therefore incorporates this lag in the index. This was evident in the RPN summaries presented.

Due to the survey changes, the GOA RPN is no longer available each year, making the previous extrapolation process impossible. The new approach is to carry forward RPN

values in missing regions until the region is surveyed. By definition this creates a lag during periods of increase and decline. For example, the last BS and AI RPNs are the highest on record and will be carried forward.

Due to these issues, using extrapolation methods to fill in missing values (which are then input as data to the assessment model) raises concerns. This has become a more significant issue with the new survey sampling scheme and potential for further changes in future. For these reasons it would be more consistent to use a modelling approach that enables missing values to be formally incorporated.

Fleets as areas (FAA) approach

An FAA model approximates a fishery with spatial structure using several fleets with selectivity curves that combine true gear selectivity and size/age specific spatial availability. It provides an accurate description of a system with complete population mixing at each time step, followed by varied spatial availability of age/size classes. The less movement there is in a given fishery, the higher the potential for biases arising from this approach. This concern needs to be balanced with the potential added complexity and information requirements of a fully spatially structured stock assessment.

Using an FAA approach, the regional RPNs can be incorporated with the missing data represented as such. This is a significant improvement over the extrapolation and carry forward approaches discussed above, however this will still create an under-estimate of the extent of an extended period of high or low recruitment (discussed further in ToR 2).

Using an FAA approach enables the spatial variations in age composition data to be represented through regional selectivity. This reduces the alternating bias that will be present in future years where age structure data will alternate between BS/AI (with higher proportional juvenile abundance) and the GoA.

The FAA approach will also enable appropriately aged animals to be removed by the fishery in each region. The TACs have historically ensured a consistent proportion of catches across the regions (making this issue less important), however recent undercatch of TAC has enabled the proportions to change (as described above).

FAA Structure

A range of FAA models and approaches were presented. The NOAA longline survey index and age composition were divided into three fleets (BS, AI and GoA) – this was included in all FAA model variants as this was one of the key drivers for using FAA. Other options examined included:

- Regional fixed gear fleets
- Different selectivity curves
- Time varying selectivity

Models were evaluated on the basis of key criteria:

- Model stability
- Data fit
- Convergence criteria
- Diagnostics including retrospective fits.

This was assessed using a five area spatial operating model to conduct both cross-testing and closed-loop simulation.

The model 26.2_FAA includes the three separate NOAA longline survey fleets with fleet specific logistic selectivity and catchability. This is the minimal complexity required to address the key elements motivating the use of a FAA model. This model provided a good fit, with good convergence and model stability. Inclusion of further complexity resulted in minimal improvement in model fit and a range of issues regarding model stability, convergence and retrospective fits.

These results indicated that 26.2_FAA incorporates the best structure for informing management decisions out of those examined. The spatial structure of 26.2_FAA and its outputs are appropriate and directly applicable to management requirements.

However, this choice limits the flexibility of the model to represent spatial complexity. For instance, the use of logistic selectivity prevents modelling a scenario where abundance of juvenile and old fish in one area are disproportionately high (e.g., AI in some years). Permitting this flexibility in the model did not notably improve fits and resulted in model stability concerns. As discussed in ToR 2, this could be indicative of issues in the data and understanding of population structure that cannot be resolved through alternative model structures.

ToR 1 Recommendations/Findings:

1. Mark-recapture analyses and spatial variations in age composition data indicate that the current single region panmictic assessment may be inappropriate.
2. Recent (and possible future) changes in the survey index make continued use of the current assessment model increasingly challenging.
3. Methods for filling missing RPN values can lead to biases, with those biased values presented to the assessment model as data. Instead, missing values should be considered as such by the assessment model, this is made possible by the FAA approach.
4. The presented FAA models provide a good balance between model complexity and realism, with clear advantages over both the single region and spatially explicit models.
5. Based on the presented results, the model 26.2_FAA provides the best trade-off between model complexity and inclusion of key underlying fishery/population

dynamics. It provides a suitable basis for operational management advice and produces the required outputs.

5. Terms of Reference 2

Highlight potential strengths and weaknesses of the FAA assessment, including model assumptions, estimates, fits to data (survey indices and age compositions along with fishery catch-at-age data), model diagnostics, and major sources of uncertainty.

a) Indicate whether the model provides the best scientific information available for estimating current stock status, projecting future abundance, and determining Overfishing Levels (OFLs) and Acceptable Biological Catches (ABCs).

Operating Model

The operating model used to test assessment models was a five-area spatial assessment model. This was parameterized based on the best available data. A key constraint here is that this model does not generate some of the characteristics of the age-frequency data that are observed and which create model fit issues.

Variants of the operating model could be used to test model robustness to alternative hypotheses. For example, inclusion of an additional population in AI that only interacts with the fishing fleet in some years. It is noted that this would be a significant piece of work and efforts may be better directed at exploring these hypotheses directly with other approaches (see ToR 3).

Survey Index Fits

The GOA Selectivity is shifted towards older animals, whilst AI and BS selectivity reaches one for all but the youngest age classes. The generalized understanding is that juveniles move to AI and BS, whilst mature fish move to the GoA. The selectivity is consistent with the first part of this generalization whilst the limitation on the selectivity curve shape prevents representing the latter.

I found that the fitted selectivity curves result in a high correlation between AI and BS RPN index fitted values. Due to the broad selectivity and long lived nature of the species, the capacity for inter-annual variability is limited. In contrast, the GOA index corresponds to a smaller number of age classes and can have greater inter-annual variability.

As GOA does not select for young age classes the AI and BS RPNs (and age structure) are the primary drivers for recent recruitment estimates. Consequently, I expected the reduction in AI and BS data to result in recent recruitment deviation estimates being closer to zero – which I confirmed. In periods of sustained high recruitment (as in recent years) this will result in an under-estimate of recent recruitment and stock status (and

vice-versa in periods of below average recruitment). Incorporating episodic recruitment in the model structure could be considered, however the information to parameterize this could be limited.

Age composition residuals

There was substantial discussion of the age composition fits and a comparison of fits across a range of model parametrizations. A key issue is the periodic presence of a plus group in the observer age composition data, appearing in 1999-2004 and 2008-2018. This plus group does not appear to the same extent in the survey age composition data. There are too few fish in intermediate age classes for the fluctuations in the plus group to have occurred through normal demographic processes.

Discussions indicated that it is likely to have arisen due to specific fishing locations visited by the observer vessel during this time. If this is the true origin, it indicates a key limitation of the assessment. One hypothesis is that there appears to be a population in which at least the old fish remain largely resident and which is only fished from time to time. Consequently, data from this population is occasionally present in observer age composition data, it is generally not sampled by the survey and an unknown amount of catch comes from this region.

Additional analysis of the observer data could be undertaken to investigate the hypothesis that spatial changes in coverage resulted in the emergence and disappearance of the plus group. If it does relate to fishing in a specific spatial area, one approach would be to remove this data from the assessment, however this would also necessitate estimating and removing the corresponding catch. Additional simulations were run at the review which removed the plus group and found that the assessment results were robust – this is reassuring for the model's suitability for support management.

Data limitations

A significant amount of effort has been applied to modelling this population. This has produced a comprehensive assessment model. This has reached the point at which the data rather than the model structure is likely the limiting factor.

Age composition data – A small proportion of the collected otoliths are aged. Others are archived. Age compositions are limited by the small sample size and much smoother distributions could be provided to the assessment model by increasing the sample size (in future and/or historic). The impacts of sample size could be guided by assessment analyses using the operating model.

Age estimates are highly uncertain, likely as fish experience limited seasonal temperature changes at their resident deep depths. As such, it is also possible that age

uncertainty is correlated with aspects such as the location and age of the fish, investigating this further could be warranted.

Survey – The scientific survey has an excellent structure that was thoroughly probed in the review. The limitation is the spatial coverage. As described here and in ToR 1, the limited coverage will result in recruitment and biomass trends lagging behind in periods of sustained high or low recruitment. This is through both the RPN and age composition data. An additional hole is that the West Aleutian Islands have not been sampled for an extended period and this is an area that is poorly understood.

Logbook catch/effort data – Historically it was possible to generate a CPUE index, however this has been discontinued due to a lack of data. I was surprised that key catch and effort logbook data is not collected. This would provide a highly valuable additional fishery abundance index, as well as allowing changes in the fishery to be examined. For example, the impact of changing gear types and changing fishing grounds within a management area. The latter may be of particular relevance in recent years as there have been large economic changes within the fishery.

Natural mortality – Natural mortality is highly constrained by the prior. Relaxing the prior resulted in model stability issues. More comprehensive sensitivity analyses (with natural mortality constrained to other values) should be conducted to quantify the impact of the prior on the OFL and ABC.

Suitability for estimating current stock status, projecting future abundance, and determining OFLs and ABCs

As described, the model has limitations, however these are largely attributable to limitations of the datasets and the required information to transition to a full spatial model is incomplete. Consequently, the FAA model provides a pragmatic balance between model complexity and realism given information limitations.

The FAA model is harder to interpret and explain, particularly as selectivity now incorporates both spatial and fleet processes. Careful consideration needs to be given on communicating this effectively. Discussion of the characteristics of different fleets may be more effective than a full explanation of the FAA analogy.

The 2027 OFL and ABCs recommended by 26.2_FAA (the preferred FAA model) are approximately 25% lower than obtained by 26.1_1Reg (the single region panmictic model). Whilst this is a notable difference it is not of an unexpected magnitude given the transition to a new assessment approach and the improvements this has entailed (see ToR 1). The information presented indicates that the 26.2_FAA provides the current best scientific information available for assessing the stock and supporting management decisions.

ToR 2 Recommendations/Findings:

1. From the presented models, 26.2_FAA provided the best compromise between parsimony and model complexity. There are some aspects of the data that it does not reproduce; however, overall, no significant issues or concerns were identified and it is appropriate for assessing the stock and supporting management decisions.
2. Results from the FAA model are harder to communicate and careful stakeholder communication will be important.
3. Natural mortality was highly constrained by the prior. Sensitivity analyses should be conducted to examine the effect on the OFL and ABC.
4. Selectivity curves were limited to logistic. This is an appropriate restriction on the basis discussed above, however it does limit the capacity of the FAA model to capture spatial aspects (such as the heightened presence of juveniles and the old plus class fish in the Aleutians). If consistent datasets for this are developed (see ToR 3) use of dome shaped selectivity should be reinvestigated.
5. The observer data should be examined spatially to determine whether spatial shifts in fishing caused the appearance of the plus group.
6. The operating model could be expanded to incorporate alternative hypotheses that reproduce observed anomalies in the age composition data.
7. The data available for the assessment model is likely to be the limiting factor for the assessment, rather than the model structure and approach. Improvements in available data would address the current weaknesses of the model.
8. Consider increasing the number of otoliths that are aged - both as a routine process in future years and potentially increasing historic sample sizes from the otolith archive. In both cases, the assessment model could be used to advise the most effective sampling regime.
9. The scientific survey is well designed but limited in spatial extent. The current spatial sampling scheme is suitable given the extremely limited resources; however, an expansion back to the previous level, or above (e.g., including WAI) would significantly improve model robustness and provide earlier indication of below/above average recruitment periods. This is a key limitation for the assessment.
10. Development of a fishery logbook including spatial data would provide an additional abundance index and enable closer consideration of changes in fishing characteristics (such as locations fished within each management area).

6. Terms of Reference 3

Provide research recommendations for improving the existing research-oriented sablefish spatial (i.e., 3 and 5 region) assessment models, especially in the context of potential future usage in a management context.

Spatial models require improved understanding of i) sufficient knowledge of stock structure to formulate an appropriate model and ii) appropriate ongoing data collection to enable the model to be fitted. Without these, it is not expected that a spatial model would provide a substantial improvement beyond the FAA model. Indeed, this has been the finding of the presented spatial model results.

Understanding stock structure

Some anomalies in the age composition data (see ToR 1 and 2) were hypothesized to be due to additional population structure not represented in the spatial structure of the data or the model structure. In the first instance, further analysis of existing datasets could provide additional guidance (e.g., examining observer data spatially as discussed in ToR 2).

Other projects / analyses that would support understanding of stock structure were discussed during the review and have the potential to inform the robust development of the spatial model. These include:

- Larval dispersal modelling to inform connectivity and explore connections to recruitment strength,
- Expanded mark-recapture tagging programs across a range of age classes and locations,
- Pop-up satellite tagging to inform movement,
- Review uncertainty in otolith ages, in particular in relation to the origin, size and age of the fish.

Ongoing Data Collection

Some potential improvements to ongoing data collection have been discussed in ToR 2. Ultimately the best spatial model structure may not neatly align with the management areas that are currently used for spatial characterization of the models. As such for both FAA and spatial models, datasets may have to be spatially restructured to alternative areas. This would apply both to future data collection as well as a need to restructure historic datasets.

Regardless of the appropriate spatial structure, the current data streams are limiting. The limitations discussed in ToR 2 are holding back the potential for further development of the FAA model and the meaningful application of a spatial model.

Expanding these programs is a key requirement for the development of a more robust assessment model.

Software engineering aspects

Bespoke stock assessment models (such as SPoRC; (Cheng et al. 2026)) enable assessment models to be highly customized to suit the fishery under assessment. However, such modelling frameworks are typically developed and maintained by a relatively small development team. This introduces a heightened risk of significant errors from the outset, or introduced as features are added. A brief review of the codebase indicated that good software engineering practices have been adopted, which will mitigate this risk, these include:

- Use of a version control system (github)
- Well organised code with good function documentation and comments
- Five complex unit tests.

I was particularly impressed with the presence of formalised unit tests – in my view these are underutilized in fisheries science. I strongly recommend that these continue to be updated to test new features and are run routinely to provide early detection of new bugs.

Individual functions are well documented, there is a good user guide and the assessment model has been published (Cheng et al. 2026). However, the assessment model would benefit from a clear design document aimed at current and future developers; in particular to ensure the codebase remains manageable as the package grows. This would be especially useful for succession planning and collaboration. This will be increasingly important as model complexity increases; as expected if spatial model development is further pursued.

ToR 3 Recommendations/Findings:

Existing data analysis

1. Investigate methods for splitting out the West Aleutian region (into separate fleets / areas).
2. Investigate temporal changes in Aleutian observer coverage and the link to changes in age composition data.
3. Examine factors that may contribute to uncertainty in otolith aging data.

Annual data collection

4. Investigate the feasibility of obtaining spatial catch and effort logbook data.
5. Ensure that age composition and survey data do not diminish below current levels (see also ToR 2) and expand these programs to support the development of a more robust assessment model.

Specific research projects

6. Support research regarding spatial stock structure including approaches such as larval dispersal modelling, pop up satellite tags and expanded mark recapture programs.

Software engineering

7. Develop and maintain a clear technical design document for SPoRC.
8. Continue to expand the SPoRC unit tests and ensure tests are conducted as a matter of routine (e.g., automated weekly).

7. Appendix 1: Bibliography

Note: A full bibliography of provided materials is not included as it is too extensive.

These are available at https://github.com/dgoethel-noaa/2026_Sablefish_CIE

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8. Appendix 2: Performance work statement

Performance Work Statement (PWS)
National Oceanic and Atmospheric Administration (NOAA)
NOAA Fisheries
Center for Independent Experts (CIE) Program
External Independent Peer Review

Alaska-wide Sablefish Stock Assessment
June 16 - 18, 2026

Background

NOAA Fisheries is mandated by the Magnuson-Stevens Fishery Conservation and Management Act, Endangered Species Act, and Marine Mammal Protection Act to conserve, protect, and manage our nation's marine living resources based upon the best scientific information available (BSIA). NOAA Fisheries science products, including scientific advice, are often controversial and may require timely scientific peer reviews that are strictly independent of all outside influences. A formal external process for independent expert reviews of the agency's scientific products and programs ensures their credibility. Therefore, external scientific peer reviews have been and continue to be essential to strengthening scientific quality assurance for fishery conservation and management actions.

Scientific peer review is defined as the organized review process where one or more qualified experts review scientific information to ensure quality and credibility. These expert(s) must conduct their peer review impartially, objectively, and without conflicts of interest. Each reviewer must also be independent from the development of the science, without influence from any position that the agency or constituent groups may have. Furthermore, the Office of Management and Budget (OMB), authorized by the Information Quality Act, requires all federal agencies to conduct peer reviews of highly influential and controversial science before dissemination, and that peer reviewers must be deemed qualified based on the OMB Peer Review Bulletin standards¹.

Sablefish (*Anoplopoma fimbria*) are a long-lived (90+ years), highly mobile, sexually dimorphic demersal species that live at depths greater than 200m (as adults). In Alaska Federal waters, sablefish are assessed and managed as a single population with the total quota apportioned across 6 management regions (Aleutian Islands, Bering Sea, Western Gulf of Alaska, Central Gulf of Alaska, Western Yakutat, and Southeast Outside) and among two fishing sectors (i.e., fixed gear and trawl gear). Since rationalization (i.e., implementation of individual fishing quotas, IFQs) occurred in the 1990s, sablefish have been one of the most valuable finfish resources in Alaska (\$124 million in ex-vessel value in 2022). Catch advice for sablefish is based on a statistical catch-at-age integrated stock assessment model fit to fishery-independent survey data and fishery-dependent catch-age data. The sablefish assessment is reviewed annually by the North Pacific Fishery Management Council's (NPFMC) Science and Statistical Committee (SSC) as part of the quota specification process, and last underwent a CIE review in 2016.

Scope

Since the last CIE review (i.e., in 2016), major changes have occurred in the sablefish resource, fishery, longline survey, and assessment model. Specifically, a series of extreme, above average recruitment events

¹ https://www.whitehouse.gov/wp-content/uploads/legacy_drupal_files/omb/memoranda/2005/m05-03.pdf

has led to rapid rebuilding of the resource and increasing quotas. Resultant landings of predominantly small, lower value fish have led to market saturation, strong reductions in landed value, and exploration of management options that loosen full retention requirements. Concomitantly, the primary directed fixed gear fleet has undergone rapid gear changes, where a predominantly hook-and-line fishery has transitioned to a longline pot fishery since 2017 (i.e., >85% of sablefish catch in the fixed gear fishery is now taken by pot gear), which has helped to reduce whale depredation.

Additionally, the fishery-independent, cooperative research-based NOAA longline survey, which targets sablefish and serves as the primary index of abundance for the assessment, underwent major survey design changes starting in 2025. Because the survey has traditionally used a cost-recovery approach, whereby survey catch is sold by the fishing company that owns the vessel performing the survey, the decline in sablefish markets prevented the survey from being undertaken in 2024 for the first time in its 30+ year history. Historically, the fixed station survey has sampled the entire Gulf of Alaska (GOA) annually, as well as either the Bering Sea (BS, odd years) or Aleutian Islands (AI, even years). Starting in 2025 a new survey design was enacted to reduce overhead and ensure a fishing company would accept the survey contract, wherein the Gulf of Alaska is now surveyed in odd years (starting in 2025), and the BS and AI are surveyed in even years (starting in 2026). Although the new survey design ensures the ability to continue the longline survey, it results in only ~50% of total stock abundance being surveyed in a given year. Moreover, it raises important challenges for the panmictic assessment in terms of how to integrate survey data that only partially samples the population being assessed.

There have also been major changes to the assessment model since the last CIE review. Most importantly, a new R-TMB based package, the [SPoRC](#) (Stochastic Population Over Regional Components) assessment model, has been adopted, which replaces the previous Automatic Differentiation Model Builder (ADMB) based assessment model. Additionally, a number of research-oriented assessment models have been developed exploring alternative spatial structures (e.g., spatially-explicit 3 and 5 region models that estimate movement among regions, as well as a spatially-implicit single region fleets-as-areas, FAA, model), all of which demonstrate promise for dealing with spatial complexities and data gaps that are not well resolved in the single region panmictic assessment. In 2026, the assessment team transitioned from the single region panmictic assessment to the FAA model, because the FAA model could better handle the new longline survey design (i.e., fit the data by region as separate fleets) and implicitly account for ontogenetic movement dynamics (i.e., through selectivity and availability assumptions). Thus, for the purpose of the CIE review, feedback on the FAA model is being requested, particularly with regard to appropriate model structure and parametrization. To help make determinations on the BSIA to assess the Alaska sablefish population, reviewers will be provided the results of closed loop simulation analyses and assessment model runs applied to the sablefish data (including model outputs and diagnostics).

The overarching goals of this review process are to:

1. Ensure the FAA stock assessment supports the development of robust advice for use by the North Pacific Fishery Management Council based on the BSIA;
2. Provide an independent external review of this stock assessment to confirm that it meets the mandates of the Magnuson-Stevens Fisheries Conservation and Management Act (MSA) and other legal requirements;
3. Understand and identify appropriate FAA model structure and fits to key data sources (indices, landings, and age compositions);
4. Identify research needed to improve the assessment and advice for fishery managers.

The specified format and contents of the individual peer review report are found in **Annex 1**. The Terms of Reference (ToRs) of the peer review are included in **Annex 2**. The tentative agenda of this in-person panel review meeting is included in **Annex 3**.

Requirements

NOAA Fisheries requires **3 reviewers** to conduct an impartial and independent peer review in accordance with this Performance Work Statement (PWS), OMB Guidelines, and the ToRs below. Modifications to this Performance Work Statement (PWS) and ToRs cannot be made during the peer review, and the Contracting Officer's Representative (COR) and the CIE contractor shall approve any modifications prior to the peer review. All ToRs must be addressed in each reviewer's report.

The CIE reviewers should have expertise in statistical sex- and age-structured integrated stock assessment models. In particular, experience developing spatially explicit and spatially implicit (i.e., fleets-as-areas) assessment models is imperative. Moreover, understanding of stock assessment good practices are required. Working knowledge of the RTMB programming language, a basic understanding of closed loop simulation, and a background in survey design and/or the use of fishery-independent survey data in stock assessments would all be beneficial.

The chair for this meeting, who is in addition to the CIE reviewer, will not be provided by the CIE. Although the chair will be participating in this review, the chair's participation (e.g., labor and travel) is not covered by this contract.

Tasks for Reviewers

The CIE reviewer shall complete the following tasks in accordance with the PWS and Schedule of Milestones and Deliverables herein.

Pre-review Background Documents: A minimum of two weeks before the peer review, the NOAA Fisheries Project Contact will send (by electronic mail or make available at an FTP site/[GitHub Repository](#)) to the CIE reviewers the necessary background information and reports for the peer review. In the case where the documents need to be mailed, the NOAA Fisheries Project Contact will consult with the CIE on where to send documents. The CIE reviewers are responsible only for the pre-review documents that are delivered to the reviewer in accordance with the PWS scheduled deadlines specified herein.

The document list is separated into 'summary/background' and 'for review' documents. The CIE reviewers are *encouraged to peruse* the 'summary/background' documents for information on data inputs, previous assessment structures and associated reviews, and ongoing research. However, the CIE reviewers *must review* all of the 'for review' documents in preparation for the peer review, as these contain the results of the primary analyses and model runs that will be discussed during the review panel.

Available documents (available from the public [sablefish CIE GitHub repository](#), as they become available) include:

For Review (available [here](#))

1. A summary document describing the fleets-as-areas assessment model, including the full model description and results, with comparison to outputs from the single region panmictic assessment as well as summary results for all sensitivity model runs;
2. A document summarizing the results of the spatially explicit closed loop simulation analysis;
3. The [2025 Alaskan sablefish stock assessment proposed model changes report](#) (Goethel and Cheng, 2025) produced for the September 2025 Joint Plan Team meeting, [presentation](#), and [GitHub repository](#), which describes recent changes to the current single region panmictic assessment including transition to RTMB and the SPoRC assessment package;
4. The spatial sablefish model publication and [model updates report](#) (Cheng et al., 2025a) for the September 2025 Joint Plan Team meeting;
5. Additional supporting documents and presentations as they become available.

Summary/Background (Available [here](#))

1. The 2024 Alaskan sablefish Stock Assessment and Fishery Evaluation (SAFE) report (Goethel and Cheng, 2024) and presentation;
2. The 2024 December SSC (p. 14-16) and November Joint Plan Team (p. 2-3) reports;
3. The 2025 AFSC Longline Survey Cruise Report (Siwicke et al., 2026) and presentation;
4. The 2025 October SSC (pp. 21-23) and September Joint Plan Team (p. 5-6) reports;
5. The SPoRC assessment platform publication (Cheng et al., 2026) and GitHub documentation;
6. Research papers on integrating density-dependent growth (Cheng et al., 2025b) and a pot gear fleet (Cheng et al., 2024a) into the sablefish assessment, simulation studies on selectivity (Cheng et al., 2024b) and sex-specific parametrization (Cheng et al., 2025c) for the sablefish assessment, and considerations for fitting sex-specific composition data (Cheng et al., 2025d) in stock assessments;
7. CIE reviewer reports from the 2016 Alaskan sablefish CIE review.

Please note that outputs of all analyses used as part of the CIE review, including input data, simulation outputs, final proposed assessment models, and all sensitivity assessment models, will be made available on the [sablefish CIE GitHub repository](#)

Panel Review Meeting: The CIE reviewers shall conduct the independent peer review in accordance with the PWS and ToRs and shall not serve in any other role unless specified herein. The CIE reviewers shall actively participate in a professional and respectful manner as members of the meeting review panel, and their peer review tasks shall be focused on the ToRs as specified herein. The NOAA Fisheries Project Contact is responsible for any meeting arrangements (e.g., conference room reservations, video conferencing logistics, etc.).

Contract Deliverables - Independent CIE Peer Review Report: Each CIE reviewer shall complete an independent peer review report in accordance with this PWS. Reviewers are not required to reach a consensus, and a summary report *will not* be produced. Each CIE reviewer shall complete their independent peer review according to the required format and content as described in **Annex 1**. Each CIE reviewer shall complete the independent peer review addressing each ToR as described in **Annex 2**. The tentative agenda of the panel review meeting is in **Annex 3**. Each CIE reviewer will deliver their reports according to the specified milestone dates.

Foreign National Security Clearance

When reviewers participate during a panel review meeting at a government facility, the NOAA Fisheries Project Contact is responsible for obtaining the Foreign National Security Clearance approval for reviewers who are non-US citizens. For this reason, the reviewers shall provide requested information (e.g., first and last name, contact information, gender, birth date, passport number, country of passport, travel dates, country of citizenship, country of current residence, and home country) to the Project Contact for the purpose of their security clearance, and this information shall be submitted at least two weeks in advance. For additional information, please see the following link: <https://www.commerce.gov/osy/programs/foreign-access-management>. All CIE reviewers must have a REAL ID-compliant form of identification to access Federally owned or leased facilities. The contractor is required to use all appropriate methods to safeguard Personally Identifiable Information (PII).

Place of Performance

The place of performance shall be Juneau, AK.

Period of Performance

The CIE reviewers' duties shall not exceed **14** days to complete all required tasks.

Schedule of Milestones and Deliverables

The contractor shall complete the tasks and deliverables in accordance with the following schedule:

Schedule	Milestones and Deliverables
Within two weeks of award	Contractor secures participation of the CIE reviewers.
At least two weeks prior to the panel review meeting	Pre-review documents are provided to the reviewers.
June 16 - 18, 2026	The reviewers participate and conduct an independent peer review during the panel review meeting.
Approximately 2 weeks later	Contractor receives draft reports.
Within 3 weeks of receiving draft reports	Contractor submits final reports to the US Government.

Applicable Performance Standards

The acceptance of the contract deliverables shall be based on three performance standards: (1) The reports shall be completed in accordance with the required formatting and content in **Annex 1**; (2) The reports shall address each ToR as specified **Annex 2**; and (3) The reports shall be delivered as specified in the schedule of milestones and deliverables.

Travel

All travel expenses shall be reimbursable in accordance with Federal Travel Regulations ([Travel resources | GSA](#)), and all contractor travel must be approved by the COR prior to the actual travel. Any travel conducted prior to the receipt of proper written authorization from the COR will be done at the Contractor's own risk and expense. International travel is authorized for this contract. Travel is not to exceed \$13,000.00.

Confidentiality and Data Privacy

This contract may require that services contractors have access to Privacy Information. Services contractors are responsible for maintaining the confidentiality of all subjects and materials and may be required to sign and adhere to a Non-disclosure Agreement (NDA).

NOAA Fisheries Project Contact:

Chris Lunsford
MESA Program Manager
NOAA Fisheries, Alaska Fisheries Science Center
17109 Pt. Lena Loop Road
Juneau, AK 99801
Chris.Lunsford@noaa.gov

Annex 1: Peer Review Report Requirements

1. The CIE independent report shall be prefaced with an Executive Summary providing a concise summary of the findings and recommendations as specified in the ToR and provide advice on whether the science reviewed is consistent with and represents the best scientific information available (BSIA).
2. The main body of the reviewer report shall consist of the following sections: Background; Description of the Individual Reviewer's Role in the Review Activities; Summary of Findings for each ToR in which the weaknesses and strengths are described; and Conclusions and Recommendations in accordance with the ToRs.
 - a. Each reviewer should describe in their own words the review activities completed during the panel review meeting, including providing a brief summary of findings as well as the science, conclusions, and recommendations.
 - b. Each reviewer should discuss their independent views on each ToR even if these were consistent with those of other panelists, and especially where there were divergent views.
 - c. Each reviewer should elaborate on any points that they feel might require further clarification.
 - d. Each independent CIE report shall be a stand-alone document through which others can understand the weaknesses and strengths of the science reviewed and shall be an independent peer review of each ToRs.
3. The reviewer report shall include the following appendices:
 - Appendix 1: Bibliography of materials provided for review.
 - Appendix 2: A copy of the CIE Performance Work Statement.
 - Appendix 3: Panel Membership or other pertinent information from the panel review meeting.

Annex 2: Terms of Reference (ToRs) for the Peer Review

Review of the Alaska-wide Sablefish Assessment

CIE reviewers are contracted to complete their independent peer review based on the ToRs. Therefore, the CIE-NOAA Fisheries review and approval process is based on whether the CIE independent reports addressed each of the ToRs.

- 1)** Review the disseminated documents and analyses, then provide feedback on whether the fleets-as-areas (FAA) assessment provides a suitable basis for operational management advice, particularly in comparison to the current single region panmictic assessment.
 - a) Provide recommendations on the most appropriate parametrization of fleet structure (number of fleets per fishery/survey, as a proxy for regions) in the FAA model.
- 2)** Highlight potential strengths and weaknesses of the FAA assessment, including model assumptions, estimates, fits to data (survey indices and age compositions along with fishery catch-at-age data), model diagnostics, and major sources of uncertainty.
 - a) Advise as to whether the model is consistent with and represents the best scientific information available (BSIA) for estimating current stock status, projecting future abundance, and determining Overfishing Levels (OFLs) and Acceptable Biological Catches (ABCs).
- 3)** Provide research recommendations for improving the existing research-oriented sablefish spatial (i.e., 3 and 5 region) assessment models, especially in the context of potential future usage in a management context.

Reviewers must provide details for the basis of their responses to the review ToRs.

Annex 3: Tentative Agenda for the CIE Review of the Alaska-wide Sablefish Assessment

North Pacific Fishery Management Council
NOAA Alaska Fisheries Science Center, Auke Bay Laboratories
17109 Pt. Lena Loop Road
Juneau, AK 99801

June 16-18, 2026

Contact: Chris Lunsford (Chris.Lunsford@noaa.gov)

Schedule

(this agenda is *DRAFT* and subject to change)

All times below are Alaska Daylight Time
Daily breaks at ~10:30 and ~15:30, Lunch 12:30-14:00
Tuesday-Thursday, 09:00 – 17:00 AKDT

Tuesday, June 16, 2026

A. Welcome and Introductions

09:00 — 09:15: Welcoming Remarks, Introductions, Logistics

Chris Lunsford

09:15 — 09:30: Announcements, Agenda/ToRs

Chair

B. Sablefish Management, Biology, Fishery, and Data Inputs (ToR 1)

09:30 — 09:50: Biology, Spatial Dynamics

Daniel Goethel

09:50 — 10:10: Sablefish Management/NPFMC FMP Requirements

Sara Cleaver

10:10 — 10:30: Fishery-Dependent Data

Matt Cheng

10:30 — 10:50: *BREAK*

10:50 — 11:15: Longline Survey Data and Recent Design Changes

Kevin Siwicke

11:30 — 12:00: Assessment History, Inputs, and Other Available Data

Daniel Goethel

12:00 — 12:30: Discussion

12:30 — 14:00: *LUNCH BREAK*

C. Assessment Model Configuration (ToR 1)

14:00 — 14:30: Current Model Development Pathway

Daniel Goethel

14:30 — 15:30: General Sablefish Assessment Parametrization

Daniel Goethel

15:30 — 15:50: *BREAK*

15:50 — 17:00: Discussion

Wednesday, June 17, 2026

A. Reconvene

09:00 — 09:15: Announcements, Agenda/ToRs

Chair

09:15 — 10:30: Discussion

10:30 — 10:50: *BREAK*

B. Spatially Implicit (Fleets-as-Areas) Model Configuration (ToRs 1)

10:50 — 12:00: Results of Spatial Simulations to Optimize FAA Structure

Matt Cheng

12:00 — 12:30: Discussion

12:30 — 14:00: *LUNCH BREAK*

C. Comparing Assessment Performance and Diagnostics (ToR 1/2)

14:00 — 14:20: Final Fleets-as-Areas Assessment Configuration

Daniel Goethel

14:20 — 15:30: Outputs/Diagnostics for FAA and Panmictic Models

Daniel Goethel

15:30 — 15:50: *BREAK*

15:50 — 17:00: Discussion

Thursday, June 18, 2026

A. Reconvene

09:00 — 09:15: Announcements, Agenda/ToRs

Chair

B. Comparing Assessment Performance and Diagnostics (ToR 1/2)

09:15 — 10:30: Discussion

10:30 — 10:50: *BREAK*

10:50 — 11:30: Results of a Spatial Closed Loop Simulation Analysis

Matt Cheng

11:30 — 12:30: Discussion

12:30 — 14:00: *LUNCH BREAK*

C. Spatial Models (ToR 3)

14:00 — 15:00: 3 and 5 Region Spatial Model Configuration/Outputs

Matt Cheng

15:00 — 15:30: Discussion

15:30 — 15:50: *BREAK*

15:50 — 17:00: Discussion and Report Writing

ADJOURN

9. Appendix 3: Panel attendance

CIE Reviewers:

Klaas Hartmann- Australia

Peter Stephenson- Australia

Sam Subbey- Norway

NOAA Attendees:

Aaron Berger (NOAA/NMFS NWFSC)

Daniel Goethel (NOAA/NMFS AFSC)

Matt Cheng (NOAA/NMFS AFSC; University of Alaska)

Chris Lunsford (NOAA/NMFS AFSC)

Kristen Omori (NOAA/NMFS AFSC)

Pete Hulson (NOAA/NMFS AFSC)

Kevin Siwicke (NOAA/NMFS AFSC)

Sara Cleaver (NPFMC)

Pat Malecha (NOAA/NMFS AFSC)

Melissa Haltuch (NOAA/NMFS AFSC)

Grant Adams (NOAA/NMFS AFSC)

Jim Ianelli (NOAA/NMFS AFSC)

Cindy Tribuzio (NOAA/NMFS AFSC)

Samara Nehemiah (ASMFC)

Beth Matta (NOAA/NMFS AFSC)

John Brogan (NOAA/NMFS AFSC)